NEXUS:
Advanced Static Analysis & Threat Intelligence

IAC Presentation

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Meet Our Team

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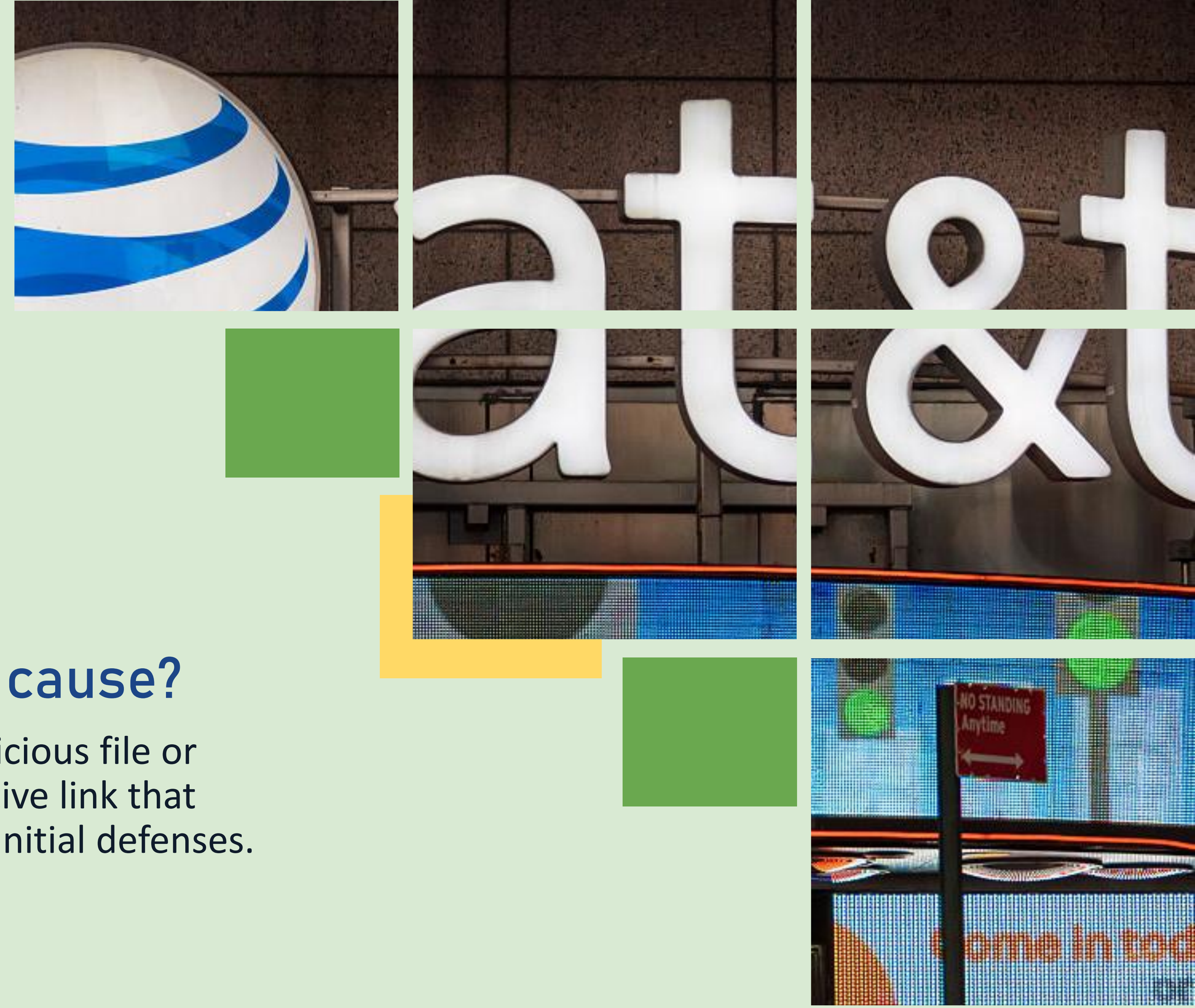
Why Threat Analysis Matters: AT&T Data Breach - March 2024

- 73 million customer records exposed
- Cost: $13 million in direct losses
- Reputation damage: immeasurable

The cause?

A suspicious file or deceptive link that bypassed initial defenses.



AGENDA

1. Problem Statement
2. Our Solution
3. System Architecture & Backend
4. Frontend Design & User Experience
5. Testing & Performance Results
6. Live Demonstration
7. Conclusion

1. PROBLEM STATEMENT

The Cybersecurity Gap

By The Numbers:	
$4.45M	Average cost of a data breach (IBM, 2023).
45%	Percentage of malware delivered through files, links, and phishing content.
92%	Percentage of organizations that rely on signature-based detection.
2-4 hrs	Manual analysis of a suspicious file can take hours depending on complexity.

The Challenge:

- Fast automated tools may miss sophisticated or novel threats.
- Thorough analysis takes too long
- Existing solutions lack transparency
- Privacy concerns with third-party uploads

Existing Tools: Limitations

AVG

✓ Fast signature-based scanning

✗ Limited explainability

✗ May miss new or obfuscated threats

✗ Limited metadata/context analysis

VirusTotal

✓ Useful quick triage

✗ Limited control / privacy

✗ third-party file submission concerns

✗ external dependency

The Gap:

Many existing tools require tradeoffs between speed, transparency, privacy, or deployment control.

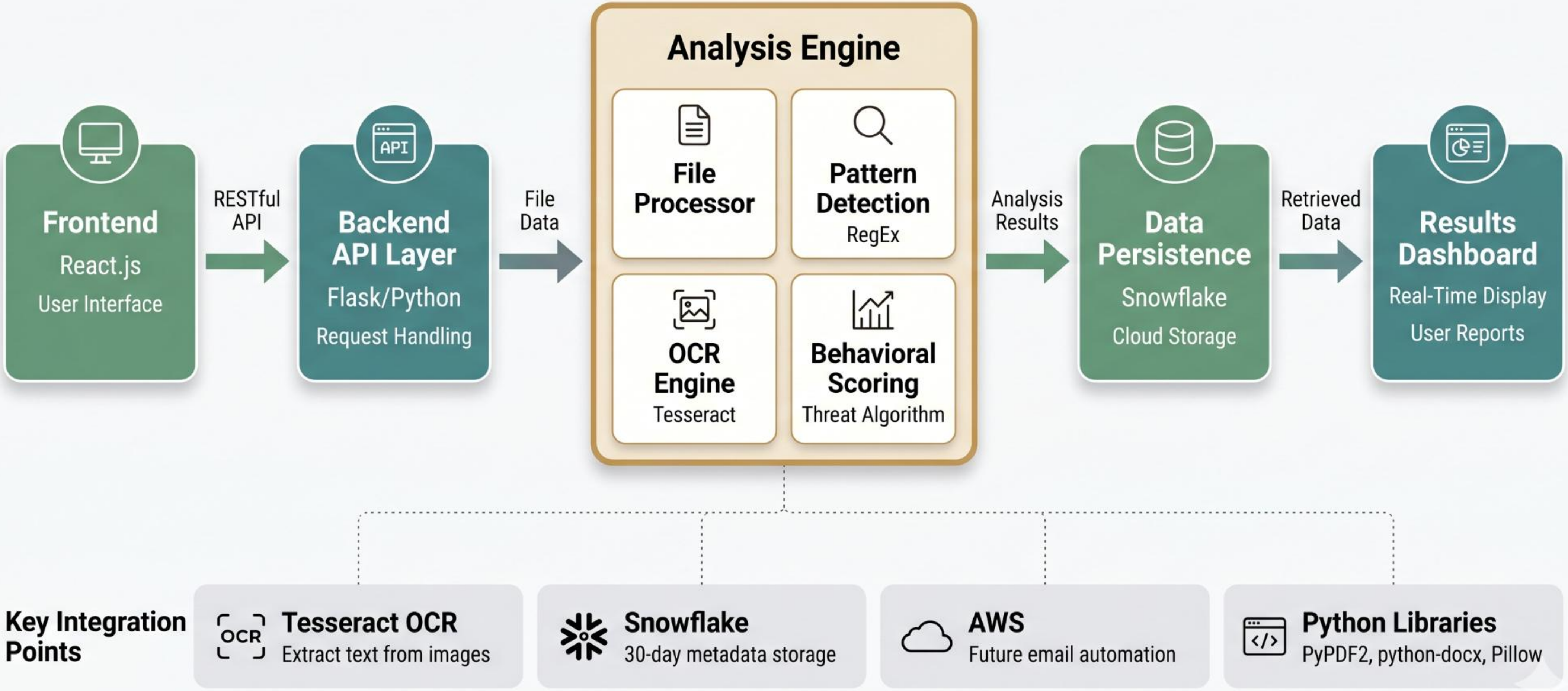
2. OUR SOLUTION: NEXUS

	Results	Privacy	Speed	Methods
Other Tools	Often limited explanation	May require external services	Often tradeoffs in speed or depth	Limited workflow flexibility
NEXUS	Why it’s malicious (ips, urls, metadata)	Local processing - Files never stored	Fast and accurate (<8 sec, 90% accuracy)	File, URL, email (3 analysis methods)

Our Distinctive Approach

3. SYSTEM ARCHITECTURE & BACKEND

Nexus Architecture: End-To-End



Teaching Maliciousness Through Indicators

NEXUS doesn't just flag threats — it explains why each indicator matters.

HOW NEXUS TEACHES

1. Detect
NEXUS scans the file for suspicious content and patterns.

2. Understand
Each indicator is analyzed in context using threat intelligence and heuristics.

3. Explain
You get clear, human-friendly explanations about why the indicator could be malicious.

4. Educate & Protect
Users learn as they review, building security awareness while staying protected.

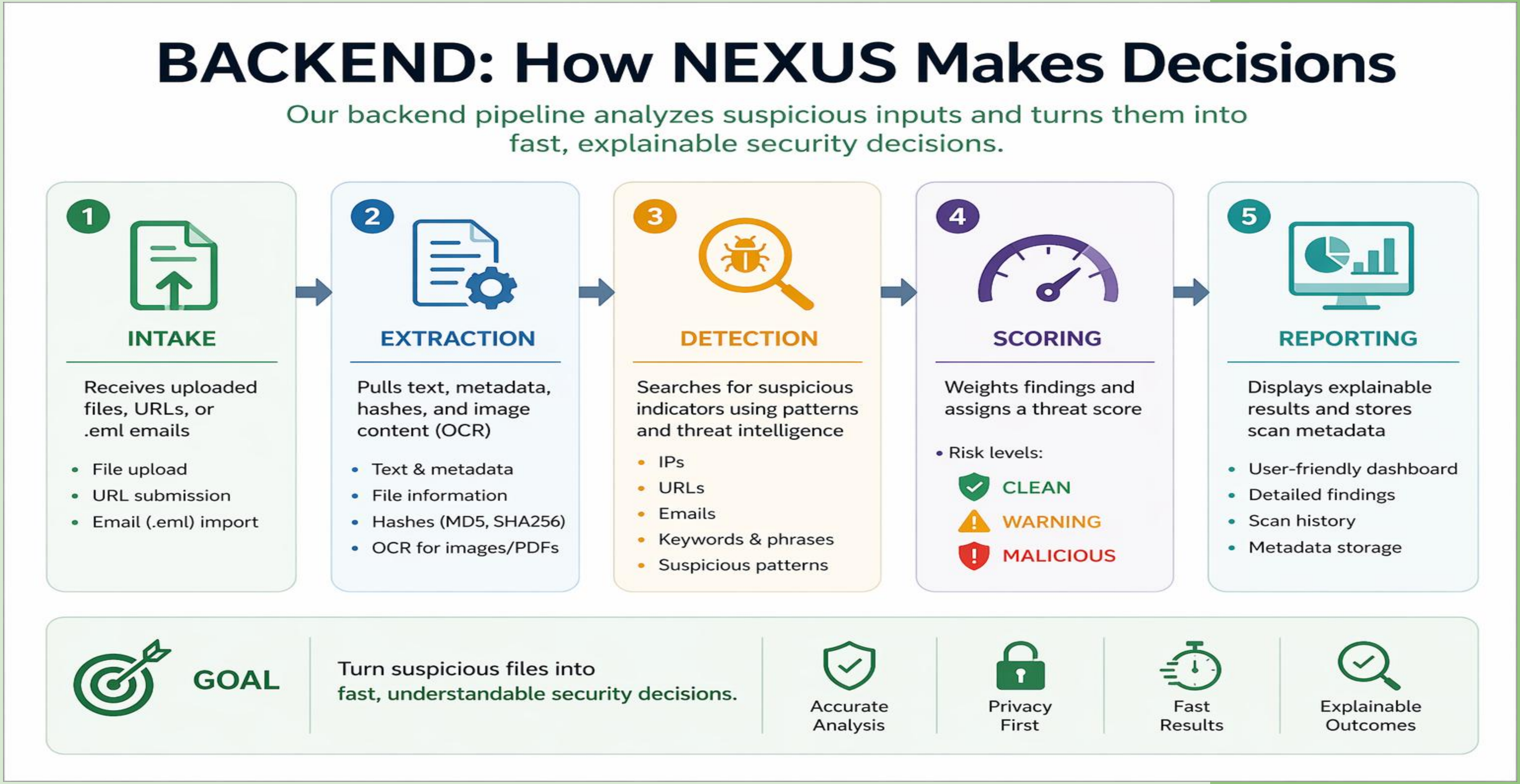
EXAMPLE INDICATORS & WHY THEY MATTER

Indicator Found	Why It Could Be Malicious	Risk Impact
Keyword: "verify account" Found in document text	Attackers use urgency and account verification themes to trick users into taking action or revealing credentials.	High
URL Detected http://secure-login-update.com	The domain is not associated with the expected organization and may lead to a phishing or malicious site.	Medium
Embedded File invoice_update.exe	Executable files can run scripts or install malware when opened.	High
OCR Text Detected Hidden text in image	Malicious actors hide instructions or links in images to evade basic text-based detection.	Medium
Suspicious Pattern Unusual formatting + urgency	Abnormal formatting combined with pressure phrases is a common social engineering technique.	Low

We don't just detect.
We help you understand.

By connecting every finding to a clear explanation, NEXUS turns every scan into a learning opportunity and a stronger defense.

Knowledge today.
Safer tomorrow.


BACKEND: How NEXUS Makes Decisions
Our backend pipeline analyzes suspicious inputs and turns them into fast, explainable security decisions.
1
INTAKE
Receives uploaded files, URLs, or .eml emails
• File upload
• URL submission
• Email (.eml) import
2
EXTRACTION
Pulls text, metadata, hashes, and image content (OCR)
• Text & metadata
• File information
• Hashes (MD5, SHA256)
• OCR for images/PDFs
3
DETECTION
Searches for suspicious indicators using patterns and threat intelligence
• IPs
• URLs
• Emails
• Keywords & phrases
• Suspicious patterns
4
SCORING
Weights findings and assigns a threat score
• Risk levels:
CLEAN
WARNING
MALICIOUS
5
REPORTING
Displays explainable results and stores scan metadata
• User-friendly dashboard
• Detailed findings
• Scan history
• Metadata storage
GOAL
Turn suspicious files into fast, understandable security decisions.
Accurate Analysis
Privacy First
Fast Results
Explainable Outcomes

4. FRONTEND DESIGN & USER EXPERIENCE

User Experience: Designed For Everyone

Simplicity

- 3-click maximum to results
- Minimal cognitive load
- No cybersecurity expertise required

Clarity

- Visual threat scoring
- Color-coded severity (Green/Amber/Red)
- Plain language explanations

Accessibility

- WCAG 2.1 AA compliance
- Screen reader compatible
- Keyboard navigation support
- High contrast ratios

Speed

- Real-time progress indicators
- Instant feedback
- Non-blocking operations

Our Design Approach

"Enterprise-grade security should feel consumer-simple."

Frontend: Technical Design

Technology	Features	Accessibility
• React.js • Responsive CSS • RESTful API • WebSocket Updates	• Drag & Drop Upload • Real-Time Progress • Threat Visualization • Export Reports	• WCAG 2.1 AA • Screen Readers • Keyboard Nav • Dark/Light Mode

Built for speed, accessibility, and ease of use.

Live System Output: Explainable Threat Results

THREAT SUMMARY

malicious_004.txt

3 indicators · Submitted via File Upload

100

THREAT SCORE

MALICIOUS

FILE TYPE	.txt
FILE SIZE	1 KB
MD5	705770217032bd5e29ed8aae2e45 2e81
SHA256	657bb4dd2d927f4f9090a114b185c dfd71aff8ba4e41f27ac92a127ca05a3 ce3
SCAN TIME	0s
SCANNED	4/24/2026, 1:53:29 AM

Run new scan →

SUSPICIOUS INDICATORS

L **URL Detected**

A URL was found in the content. A link alone is not always malicious, but it becomes more suspicious when paired with phishing terms, suspicious domains, dangerous extensions, or unusual ports.

M **Suspicious Domain Detected**

The URL uses a top-level domain that is commonly abused in phishing or malware campaigns. Attackers often choose cheap or easy-to-register domains to quickly deploy malicious links.

H **Non-Standard Network Port**

The URL uses a port other than common web ports like 80 or 443. Unusual ports can indicate command-and-control traffic, suspicious hosting, or attempts to evade normal web filtering.

M **Suspicious URL Path Keyword**

The URL contains words often associated with phishing or social engineering, such as login, verify, update, reset, or payment. These terms are commonly used to trick users into trusting malicious pages.

H **Dangerous File Extension in URL**

The link appears to point to a potentially dangerous downloadable file such as an executable, script, or macro-enabled document. These file types are often used to deliver malware.

M **Multiple URLs Detected**

The content contains multiple URLs. A high concentration of links can be suspicious in phishing documents, redirect chains, or malicious lure files.

M **Phishing Language Detected**

The content contains words or phrases commonly used in phishing campaigns, such as urgent, verify, confirm, or reset password. These terms are meant to pressure users into acting quickly.

L **Email Address Found**

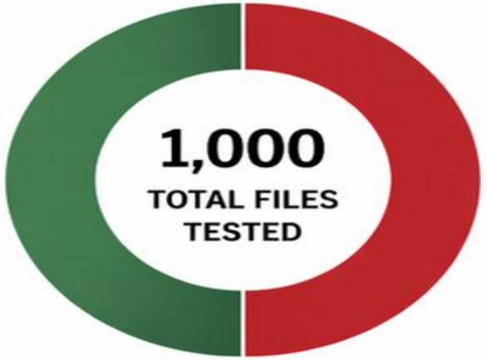
5. TESTING & PERFORMANCE RESULTS

TESTING DATASET OVERVIEW

We evaluated NEXUS using a balanced and diverse dataset to ensure reliable and robust results.

DATASET COMPOSITION

500
CLEAN FILES
Legitimate, safe files with no malicious content



500
MALICIOUS FILES
Malware samples and files containing known threat indicators

FILE TYPE DISTRIBUTION

.DOCX	.PDF	.TXT	.PNG
250 files (25%)	250 files (25%)	250 files (25%)	250 files (25%)

WHAT THIS ENSURES

Balanced Evaluation
Equal mix of clean and malicious files to measure accuracy and reliability.

Format Diversity
Multiple file types to test real-world scenarios and content variations.

Robust Validation
Ensures NEXUS can detect threats across different file formats and content.

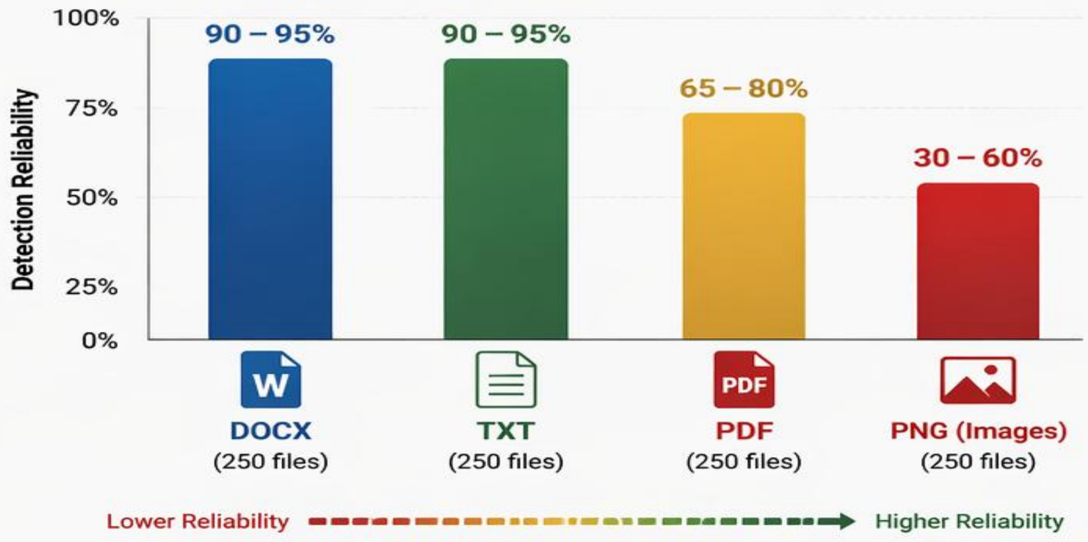
Reliable Results
Large dataset strengthens confidence in system performance.

Total: 1,000 Files Tested | 500 Clean | 500 Malicious | 4 File Types

TESTING RESULTS BY FILE TYPE

NEXUS was evaluated on a dataset of **1,000 files (500 clean, 500 malicious)** across 4 file types to measure detection performance and reliability.

DETECTION RELIABILITY BY FILE TYPE



PERFORMANCE SUMMARY BY FILE TYPE

File Type	Detection Reliability	Reason
DOCX	90 – 95% High	Text is directly readable. Consistent pattern matching and indicator detection.
TXT	90 – 95% High	Plain text format makes extraction and analysis very reliable.
PDF	65 – 80% Moderate – High	Performance depends on how text is embedded or scanned within the PDF.
PNG (Images)	30 – 60% Variable	OCR accuracy depends on image quality, font, contrast, and layout complexity.

KEY TAKEAWAYS

Stronger on Text-Based Files
DOCX and TXT files show the highest reliability due to direct and accurate text extraction.

PDFs Are Moderately Reliable
Detection depends on structure and whether the PDF contains selectable text or scanned content.

Images Are the Biggest Challenge
OCR limitations lead to lower and more variable performance on image-based files such as PNG.

These results highlight the strengths of NEXUS on text-based content and the areas where OCR improvements and advanced preprocessing can further enhance detection in image and scanned documents.

BACKEND THREAT SCORE LOGIC

NEXUS uses a rule-based scoring engine that assigns weighted points to suspicious indicators and additional points for high-risk combinations. The final score is normalized to a range of 0–100.

1. INDIVIDUAL INDICATOR WEIGHTS

INDICATOR	WEIGHT (POINTS)
URL Detected	+5
Multiple URLs (2 or more)	+10
Email Address Found	+5
Phishing Language / Keywords (e.g., urgent, verify, account, login)	+15
Suspicious Domain (low reputation / blacklisted)	+20
Dangerous File Extension in URL (e.g., .exe, .zip, .scr, .js)	+20
Non-Standard or Suspicious Port (e.g., :8080, :4444, :8888)	+15

2. COMBINATION BONUSES

COMBINED INDICATORS	BONUS (POINTS)
URL + Email (appears together)	+15
URL + Phishing Language (appears together)	+20
Multiple Indicator Types (3 or more different types)	+25

SCORING NOTES

- Points from all triggered indicators are added together.
- Combination bonuses are applied on top of individual weights.
- Final score is capped at 100 points.

3. FINAL SCORE INTERPRETATION (0–100)

0 – 30	31 – 69	70 – 100
CLEAN / LOW RISK	**WARNING / MODERATE RISK**	**MALICIOUS / HIGH RISK**
No significant threats detected.	Suspicious content detected. Exercise caution.	Highly suspicious content. Immediate attention recommended.

This scoring model is transparent, explainable, and easy to tune as new threat intelligence and patterns are discovered.

6. LIVE DEMONSTRATION

Analysis Method	What We'll Demonstrate
File Upload (Primary Demo)	•Upload malicious PDF/TXT sample •Upload PNG image Sample •Threat score + Indicator Explanations
URL Analysis (Brief Demo)	•Submit Sample URL •Indicator Review •Risk scoring workflow
Email File (.eml) Analysis (Brief Demo)	•Upload .eml sample •Sender/body extraction •Phishing indicator detection
Results Dashboard	•Threat score explanation •Show why item was flagged •User-friendly findings display

Live Demo

7. CONCLUSION

Conclusion & Thank You

NEXUS: Explainable Threat Analysis

- Detects suspicious indicators in files, URLs, and email samples
- Uses OCR for image-based content
- Provides threat scores with clear explanations
- Helps users make faster, smarter security decisions

In conclusion, NEXUS demonstrates how accessible, explainable security tools can help users identify suspicious content more efficiently.

Thank you so much for you time!

Any Questions?